

**K. J. Somaiya College of Engineering, Mumbai-77**  
(Autonomous College Affiliated to University of Mumbai)

**End Semester Exam**  
Nov – Dec 2019

**Max. Marks: 50**

Class: TY

Name of the Course: Theory of Computer Science

Course Code: UCEC503

**Duration: 2Hours**

Semester: V

Branch: COMP

**Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

PWD

**Section: A**

| Question No. |                                                                                                                                                                                                                                  | Max. Marks   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|---|---|----|---|---|---|---|---|---|---|---|---|---|--|
| Q1-a         | Design FA that reads strings made up of letters in the word 'CHARIOT' and accepts only those strings that contain the words 'CAT' as a substring.                                                                                |              |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|              | Logic                                                                                                                                                                                                                            | [2M]         |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|              | Transition Table                                                                                                                                                                                                                 | [2M]         |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|              | Transition Diagram                                                                                                                                                                                                               | [2M]         |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|              | Tuple                                                                                                                                                                                                                            | [2M]         |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|              | Simulation                                                                                                                                                                                                                       | [2M]         |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|              | OR                                                                                                                                                                                                                               |              |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|              | Convert the NFA $[\{p, q, r, s\}, \{0, 1\}, \delta, p, \{s\}]$ to its equivalent DFA, where the state transition function $\delta$ is as shown in following table.                                                               |              |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|              | <table border="1"><tr><td>Q\S</td><td>0</td><td>1</td></tr><tr><td>p</td><td>p,</td><td>p</td></tr><tr><td>q</td><td>r</td><td>r</td></tr><tr><td>r</td><td>s</td><td>-</td></tr><tr><td>s</td><td>s</td><td>s</td></tr></table> | Q\S          | 0 | 1 | p | p, | p | q | r | r | r | s | - | s | s | s |  |
|              | Q\S                                                                                                                                                                                                                              | 0            | 1 |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
| p            | p,                                                                                                                                                                                                                               | p            |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
| q            | r                                                                                                                                                                                                                                | r            |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
| r            | s                                                                                                                                                                                                                                | -            |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
| s            | s                                                                                                                                                                                                                                | s            |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|              |                                                                                                                                                                                                                                  | [4M]         |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|              | Conversion process                                                                                                                                                                                                               | [2M]         |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|              | Final Table of DFA                                                                                                                                                                                                               | [2M]         |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|              | Transition Diagram of DFA                                                                                                                                                                                                        | [2M]         |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
|              | Tuple                                                                                                                                                                                                                            |              |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |
| Q1-b         | Stage Pumping Lemma of regular languages. Prove that the following language is non-regular using pumping lemma. $L=\{a^nb^{n+1} n>0\}$                                                                                           | [5M]<br>[5M] |   |   |   |    |   |   |   |   |   |   |   |   |   |   |  |

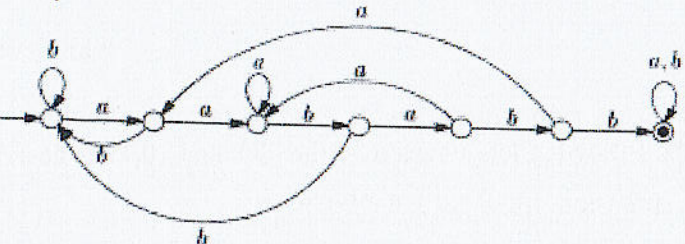


|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Q2-a | <p>Construct NFA with <math>\epsilon</math> moves that recognizes the regular expression:<br/> <math>[1(00)^*1+01^*0]^*</math> Convert this NFA to DFA</p> <p style="text-align: center;"><b>OR</b></p> <p>Construct NFA with <math>\epsilon</math> moves for the language over <math>\{0, 1\}</math> having all strings such that the third symbol from right end is 1. Convert this NFA to DFA</p> <p>Original NFA [2M]</p> <p>DFA Diagram [4M]</p> <p>DFA transition table [2M]</p> <p>DFA tuple [2M]</p>   |   |
| Q2-b | <p>Construct moore and mealy machine for the following:- for input form <math>\Sigma^*</math>, where <math>\Sigma=\{0,1\}</math>, if input ends in '101' then output should be A, if input ends in '110' output should be B, otherwise output should be C.</p> <p>Moore Machine Transition table [2M]</p> <p>Moore Machine Transition diagram [2M]</p> <p>Moore Machine Tuple [1M]</p> <p>Mealy Machine Transition table [2M]</p> <p>Mealy Machine Transition diagram [2M]</p> <p>Mealy Machine Tuple [1M]</p> |   |
| Q3   | Answer following questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Q3-a | <p>Which regular expression best describes the language accepted by the non-deterministic automaton below?</p> <div style="text-align: center;"> </div> <p>(A) <math>(a + b)^* a(a + b)b</math></p> <p>(B) <math>(abb)^*</math></p> <p>(C) <math>(a + b)^* a(a + b)^* b(a + b)^*</math></p> <p>(D) <math>(a + b)^*</math></p>                                                                                                                                                                                  | 1 |



|      |                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Q3-b | <p>Given the language <math>L = \{ab, aa, baa\}</math>, which of the following strings are in <math>L^*</math>?</p> <p>1) abaabaaabaa<br/> 2) aaaabaaaa<br/> 3) baaaaabaaaab<br/> 4) baaaaabaa</p> <p>(A) 1, 2 and 3<br/> (B) 2, 3 and 4<br/> (C) 1, 2 and 4<br/> (D) 1, 3 and 4</p>                                                                                                                                            | 1 |
| Q3-c | <p>Which one of the following languages over the alphabet <math>\{0,1\}</math> is described by the regular expression: <math>(0+1)^*0(0+1)^*0(0+1)^*</math>?</p> <p>(A) The set of all strings containing the substring 00.<br/> (B) The set of all strings containing at most two 0's.<br/> (C) The set of all strings containing at least two 0's.<br/> (D) The set of all strings that begin and end with either 0 or 1.</p> | 1 |
| Q3-d | <p>Which of the following statements is false?</p> <p>(A) Every NFA can be converted to an equivalent DFA<br/> (B) Every non-deterministic Turing machine can be converted to an equivalent deterministic Turing machine<br/> (C) Every regular language is also a context-free language<br/> (D) Every subset of a recursively enumerable set is recursive</p>                                                                 | 1 |
| Q3-e | <p>The smallest finite automation which accepts the language <math>\{x \mid \text{length of } x \text{ is divisible by } 3\}</math> has :</p> <p>(A) 2 states<br/> (B) 3 states<br/> (C) 4 states<br/> (D) 5 states</p>                                                                                                                                                                                                         | 1 |



|      |                                                                                                                                                                                                                                                                                                                                                                                                        |   |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Q3-f | <p>Let L denotes the language generated by the grammar <math>S \rightarrow 0S0 \mid 00</math></p> <p>Which of the following is true?</p> <p>(A) <math>L = 0^+</math></p> <p>(B) L is regular but not <math>0^+</math></p> <p>(C) L is context free but not regular</p> <p>(D) L is not context free</p>                                                                                                | 1 |
| Q3-g | <p>Consider the following Deterministic Finite Automata</p>  <p>Which of the following is true?</p> <p>(A) It only accepts strings with prefix as "aababb"</p> <p>(B) It only accepts strings with substring as "aababb"</p> <p>(C) It only accepts strings with suffix as "aababb"</p> <p>(D) None of the above</p> | 1 |
| Q3-h | <p>Following context free grammar</p> $S \rightarrow aB \mid bA$ $A \rightarrow b \mid aS \mid bAA$ $B \rightarrow b \mid bS \mid aBB$ <p>generates strings of terminals that have</p> <p>(A) equal number of a's and b's</p> <p>(B) odd number of a's and odd number b's</p> <p>(C) even number of a's and even number of b's</p> <p>(D) odd number of a's and even number of a's</p>                 | 1 |

|      |                                                                                                                                                                                                                                                                                                                                                                   |   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Q3-i | <p>Which one of the following regular expressions is NOT equivalent to the regular expression <math>(a + b + c)^*</math>?</p> <p>(A) <math>(a^* + b^* + c^*)^*</math><br/> (B) <math>(a^*b^*c^*)^*</math><br/> (C) <math>((ab)^* + c^*)^*</math><br/> (D) <math>(a^*b^* + c^*)^*</math></p>                                                                       | 1 |
| Q3-j | <p>Consider the context-free grammar below, S is the start symbol, a and b are terminals, and <math>\epsilon</math> denotes the empty string</p> <p><math>S \rightarrow aSa \mid bSb \mid a \mid b \mid \epsilon</math></p> <p>Which of the following strings is NOT generated by the grammar?</p> <p>(A) aaaa<br/> (B) baba<br/> (C) abba<br/> (D) babaaabab</p> | 1 |



02/12/2019 (E)

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**End Semester Exam**  
Nov – Dec 2019

PWD

Max. Marks: 50

Class: TY *B-Tech*

Name of the Course: Theory of Computer Science

Course Code: UCEC503

Duration: 2Hours

Semester: V

Branch: COMP

**Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

**Section: B**

| Question No. |                                                                                                                                  | Max. Marks |
|--------------|----------------------------------------------------------------------------------------------------------------------------------|------------|
| Q1-a         | Design a push down automata which accepts the language denoted by the following grammar.<br>$S \rightarrow aSa bSb a b \epsilon$ |            |
|              | <b>OR</b>                                                                                                                        |            |
|              | Construct a PDA that accepts the language $L = \{a^n b^n   n \geq 0\}$                                                           |            |
|              | Logic                                                                                                                            | [2M]       |
|              | Transition Table                                                                                                                 | [2M]       |
|              | Transition Diagram                                                                                                               | [2M]       |
| Q1-b         | Tuple                                                                                                                            | [2M]       |
|              | Simulation                                                                                                                       | [2M]       |
|              | Convert following grammar into CNF.<br>$S \rightarrow ABA$<br>$A \rightarrow aA \epsilon$<br>$B \rightarrow bB \epsilon$         |            |
|              | Simplification of grammar                                                                                                        | [5M]       |
|              | Conversion                                                                                                                       | [5M]       |



|      |                                                                                                                                                                                                                                                                                                                                                                              |                                                             |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Q2-a | <p>Consider the following grammar <math>G = (V, T, P, S)</math>, <math>V = \{S, X\}</math>, <math>T = \{a, b\}</math> and productions <math>P</math> are</p> <p><math>S \rightarrow aSb \mid aX</math></p> <p><math>X \rightarrow Xa \mid Sa \mid a</math></p> <p>Convert this grammar in Greibach Normal Form (GNF).</p> <p>Simplification of grammar</p> <p>Conversion</p> | <p>[5M]</p> <p>[5M]</p>                                     |
| Q2-b | <p>Design a Turing machine to find 2's complement of a given binary number.</p> <p style="text-align: center;"><b>OR</b></p> <p>Design a Turing machine that recognizes words of the form <math>0^n 1^n</math> for <math>n \geq 0</math>.</p> <p>Logic</p> <p>Transition Table</p> <p>Transition Diagram</p> <p>Tuple</p> <p>Simulation</p>                                  | <p>[2M]</p> <p>[2M]</p> <p>[2M]</p> <p>[2M]</p> <p>[2M]</p> |
| Q3   | Answer following questions                                                                                                                                                                                                                                                                                                                                                   |                                                             |
| Q3-a | <p>What is the language generated by this regular expression ?</p> <p><math>(b + \epsilon)(a + ab)^*</math></p> <p>(A) Always starts with b</p> <p>(B) Can have any number of ba and ab</p> <p>(C) Can not have 2 b's together.</p> <p>(D) Starts and end with same symbol</p>                                                                                               | 1                                                           |
| Q3-b | <p>Two finite state machines are said to be equivalent if they:</p> <p>(A) Have the same number of edges</p> <p>(B) Have the same number of states</p> <p>(C) Recognize the same set of tokens</p> <p>(D) Have the same number of states and edges</p>                                                                                                                       | 1                                                           |



|      |                                                                                                                                                                                                                                                                                                                                               |   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Q3-c | <p>If all the production rules have single non - terminal symbol on the left side, the grammar defined is :</p> <p>(A) context free grammar</p> <p>(B) context sensitive grammar</p> <p>(C) unrestricted grammar</p> <p>(D) phrase grammar</p>                                                                                                | 1 |
| Q3-d | <p>Set of regular languages over a given alphabet set is closed under</p> <p>(A) union</p> <p>(B) complementation</p> <p>(C) intersection</p> <p>(D) All of these</p>                                                                                                                                                                         | 1 |
| Q3-e | <p>A given grammar is called ambiguous if</p> <p>(A) two or more productions have the same non-terminal on the left hand side</p> <p>(B) a derivation tree has more than one associated sentence</p> <p>(C) there is a sentence with more than one derivation tree corresponding to it</p> <p>(D) brackets are not present in the grammar</p> | 1 |
| Q3-f | <p>In mealy machine, the O/P depends upon?</p> <p>a) State</p> <p>b) Previous State</p> <p>c) State and Input</p> <p>d) Only Input</p>                                                                                                                                                                                                        | 1 |
| Q3-g | <p>The Grammar can be defined as: <math>G=(V, \Sigma, p, S)</math></p> <p>In the given definition, what does S represents?</p> <p>a) Accepting State</p> <p>b) Starting Variable</p> <p>c) Sensitive Grammar</p> <p>d) None of these</p>                                                                                                      | 1 |



|             |                                                                                                                                                                                                                                                                              |   |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Q3-h</b> | <p>The production of the form <math>A \rightarrow B</math> , where A and B are non-terminals is called</p> <ul style="list-style-type: none"> <li>a) Null production</li> <li>b) Unit production</li> <li>c) Greibach Normal Form</li> <li>d) Chomsky Normal Form</li> </ul> | 1 |
| <b>Q3-i</b> | <p>A Turing machine operates over:</p> <ul style="list-style-type: none"> <li>a) finite memory tape</li> <li>b) infinite memory tape</li> <li>c) depends on the algorithm</li> <li>d) none of the mentioned</li> </ul>                                                       | 1 |
| <b>Q3-j</b> | <p>In moore machine, the O/P depends upon?</p> <ul style="list-style-type: none"> <li>a) State</li> <li>b) Previous State</li> <li>c) State and Input</li> <li>d) Only Input</li> </ul>                                                                                      | 1 |



2.12.2019(E)

**K. J. Somaiya College of Engineering, Mumbai-77**  
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**End Semester Exam**

Nov – Dec 2019

**Max. Marks: 20**

**Class: TY (B. Tech)**

**Name of the Course: Theory of Computer Science**

**Course Code: UCEC503**

**Duration: 3Hours**

**Semester: V**

**Branch: COMP**

**Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

| Question No. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Max. Marks |   |   |   |      |   |   |   |   |   |   |   |   |   |   |    |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---|---|---|------|---|---|---|---|---|---|---|---|---|---|----|
| Q1(a)        | <p>Design FA that reads strings made up of letters in the word 'CHARIOT' and accepts only those strings that contain the words 'CAT' as a substring.</p> <p align="center"><b>OR</b></p> <p>Convert the NFA <math>[\{p, q, r, s\}, \{0, 1\}, \delta, p, \{s\}]</math> to its equivalent DFA, where the state transition function <math>\delta</math> is as shown in following table.</p> <table border="1"> <tr> <td>Q\S</td><td>0</td><td>1</td></tr> <tr> <td>p</td><td>p, q</td><td>p</td></tr> <tr> <td>q</td><td>r</td><td>r</td></tr> <tr> <td>r</td><td>s</td><td>-</td></tr> <tr> <td>s</td><td>s</td><td>s</td></tr> </table> | Q\S        | 0 | 1 | p | p, q | p | q | r | r | r | s | - | s | s | s | 10 |
| Q\S          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1          |   |   |   |      |   |   |   |   |   |   |   |   |   |   |    |
| p            | p, q                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | p          |   |   |   |      |   |   |   |   |   |   |   |   |   |   |    |
| q            | r                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | r          |   |   |   |      |   |   |   |   |   |   |   |   |   |   |    |
| r            | s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -          |   |   |   |      |   |   |   |   |   |   |   |   |   |   |    |
| s            | s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | s          |   |   |   |      |   |   |   |   |   |   |   |   |   |   |    |
| Q1(b)        | <p>Prove that the following language is non-regular using pumping lemma.</p> <ol style="list-style-type: none"> <li>1. <math>L = \{a^n b^{n+1}   n &gt; 0\}</math></li> <li>2. <math>L = \{ww   w \in \{0,1\}^*\}</math></li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                    | 10         |   |   |   |      |   |   |   |   |   |   |   |   |   |   |    |
| Q2(a)        | <p>Construct NFA with <math>\epsilon</math> moves that recognizes the regular expression: <math>[1(00)^*1+01^*0]^*</math> Convert this NFA to DFA</p> <p align="center"><b>OR</b></p> <p>Construct NFA with <math>\epsilon</math> moves for the language over <math>\{0, 1\}</math> having all strings such that the third symbol from right end is 1. Convert this NFA to DFA</p>                                                                                                                                                                                                                                                     | 10         |   |   |   |      |   |   |   |   |   |   |   |   |   |   |    |



|       |                                                                                                                                                                                                                                     |    |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Q2(b) | Construct moore and mealy machine for the following:- for input form $\Sigma^*$ , where $\Sigma=\{0,1\}$ , if input ends in '101' then output should be A, if input ends in '110' output should be B, otherwise output should be C: | 10 |
| Q3(a) | Design a push down automata which accepts the language denoted by the following grammar.<br>$S \rightarrow aSa   bSb   a   b   \epsilon$<br><b>OR</b><br>Construct a PDA that accepts the language $L = \{a^n b^n   n \geq 0\}$     | 10 |
| Q3(b) | Convert following grammar into CNF.<br>$S \rightarrow ABA$<br>$A \rightarrow aA   \epsilon$<br>$B \rightarrow bB   \epsilon$                                                                                                        | 10 |
| Q4(a) | Consider the following grammar $G = (V, T, P, S)$ , $V = \{S, X\}$ , $T = \{a, b\}$ and productions P are<br>$S \rightarrow aSb   aX$<br>$X \rightarrow Xa   Sa   a$<br>Convert this grammar in Greibach Normal Form (GNF).         | 10 |
| Q4(b) | State pumping lemma of context free languages. Prove that the following language is not context free, using pumping lemma.<br>$L = \{x^n y^n z^n   n \geq 1\}$                                                                      | 10 |
| Q5(a) | Design a Turing machine to find 2's complement of a given binary number.<br><b>OR</b><br>Design a Turing machine that recognizes words of the form $0^n 1^n$ for $n \geq 0$ .                                                       | 10 |
| Q5(b) | Write short notes on :<br>1. Recursive and Recursively Enumerable Languages.<br>2. Post correspondence problem.                                                                                                                     | 10 |